

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.
11. Norman S. Nise, *Control Systems Engineering*, 7th edition, Wiley, 2015.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 40****Topics covered :**

1. Construction of Fundamental set of solutions

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- Consider the homogeneous system:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad (1)$$

where $\mathbf{A}$ is a real $n \times n$ matrix. Suppose $\mathbf{A}$ has a repeated eigenvalue α , meaning α appears multiple times as a root of the characteristic equation. The **algebraic multiplicity** of an eigenvalue is the number of times it appears as a root of the characteristic polynomial. The **geometric multiplicity** of an eigenvalue is the dimension of the nullspace of $(\mathbf{A} - \alpha\mathbf{I})$, i.e., the number of linearly independent eigenvectors associated with α .

- In general:
 - For each eigenvalue, the geometric multiplicity is always less than or equal to its algebraic multiplicity.
 - Moreover, the sum of geometric multiplicities of all eigenvalues is at most n .
 - If the sum equals n , then the matrix $\mathbf{A}$ is diagonalizable, and we can construct a complete set of n linearly independent eigenvectors.
 - If the sum is strictly less than n , the matrix is not diagonalizable, and generalized eigenvectors must be used to complete the fundamental set of solutions.
- **Special Case:** Suppose α has algebraic multiplicity 2 but only one linearly independent eigenvector. Then:
 - The geometric multiplicity is 1, while the algebraic multiplicity is 2.
 - Only one eigenvector is available; an additional generalized solution must be constructed to complete the fundamental set.
- Suppose we are given one eigenvector $\mathbf{v}_1$ corresponding to the repeated eigenvalue λ_1 (for simplicity, we denote the repeated eigenvalue by λ_1 throughout the discussion.). Thus, we have one solution:

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda_1 t}. \quad (2)$$

To find a second, linearly independent solution, we propose a trial solution of the form:

$$\mathbf{x}_2(t) = t\mathbf{v}_1 e^{\lambda_1 t} + \mathbf{v}_2 e^{\lambda_1 t}, \quad (3)$$

where $\mathbf{v}_2$ is a constant vector to be determined. Substituting $\mathbf{x}_2(t)$ into the differential equation $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ gives

$$\begin{aligned}\frac{d\mathbf{x}_2}{dt} &= \frac{d}{dt} (t\mathbf{v}_1 e^{\lambda_1 t} + \mathbf{v}_2 e^{\lambda_1 t}) \\ &= \mathbf{v}_1 e^{\lambda_1 t} + t\mathbf{v}_1 \lambda_1 e^{\lambda_1 t} + \mathbf{v}_2 \lambda_1 e^{\lambda_1 t} \\ &= e^{\lambda_1 t} (\mathbf{v}_1 + \lambda_1 t\mathbf{v}_1 + \lambda_1 \mathbf{v}_2).\end{aligned}\tag{4}$$

and

$$\begin{aligned}\mathbf{A}\mathbf{x}_2(t) &= \mathbf{A} (t\mathbf{v}_1 e^{\lambda_1 t} + \mathbf{v}_2 e^{\lambda_1 t}) \\ &= e^{\lambda_1 t} (t\mathbf{A}\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2).\end{aligned}\tag{5}$$

Since $\mathbf{v}_1$ is an eigenvector corresponding to λ_1 , we know $\mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$, hence:

$$\mathbf{A}\mathbf{x}_2(t) = e^{\lambda_1 t} (\lambda_1 t\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2).$$

Now, equating the two sides (since $\frac{d\mathbf{x}_2}{dt} = \mathbf{A}\mathbf{x}_2$) results in

$$\mathbf{v}_1 + \lambda_1 \mathbf{v}_2 = \mathbf{A}\mathbf{v}_2.\tag{6}$$

Rearranging results in

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_2 = \mathbf{v}_1.\tag{7}$$

To construct the second solution, find a vector $\mathbf{v}_2$ satisfying the equation:

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_2 = \mathbf{v}_1.\tag{8}$$

The vector $\mathbf{v}_2$ is called a [generalized eigenvector](#). We have constructed two linearly independent solutions corresponding to the repeated eigenvalue λ_1 :

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda_1 t},\tag{9}$$

$$\mathbf{x}_2(t) = (t\mathbf{v}_1 + \mathbf{v}_2) e^{\lambda_1 t}.\tag{10}$$

Therefore, the general solution to the system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ is a linear combination of these two solutions:

$$\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t),\tag{11}$$

where c_1 and c_2 are arbitrary real constants determined by the initial conditions. Substituting the explicit expressions for $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$, the general solution can be written as:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 (t\mathbf{v}_1 + \mathbf{v}_2) e^{\lambda_1 t}.\tag{12}$$

Simplifying further

$$\mathbf{x}(t) = e^{\lambda_1 t} [(c_1 + c_2 t)\mathbf{v}_1 + c_2 \mathbf{v}_2]. \quad (13)$$

This form highlights that the solution involves not just $e^{\lambda_1 t}$ but also a polynomial term $te^{\lambda_1 t}$, arising due to the lack of a sufficient number of linearly independent eigenvectors.

- Suppose the repeated eigenvalue λ_1 has an algebraic multiplicity greater than 2, say k . In general, if there are not enough linearly independent eigenvectors to form the fundamental set of solutions, we proceed by constructing a chain of generalized eigenvectors. We seek vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ satisfying the following system of equations:

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_1 = \mathbf{0}, \quad (\text{standard eigenvector}) \quad (14)$$

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_2 = \mathbf{v}_1, \quad (15)$$

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_3 = \mathbf{v}_2, \quad (16)$$

$$\vdots \quad (17)$$

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_k = \mathbf{v}_{k-1}. \quad (18)$$

where

- $\mathbf{v}_1$ is the ordinary eigenvector.
- $\mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_k$ are generalized eigenvectors of higher order.

Correspondingly, the solutions to $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ are:

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda_1 t}, \quad (19)$$

$$\mathbf{x}_2(t) = (t\mathbf{v}_1 + \mathbf{v}_2) e^{\lambda_1 t}, \quad (20)$$

$$\mathbf{x}_3(t) = \left(\frac{t^2}{2} \mathbf{v}_1 + t\mathbf{v}_2 + \mathbf{v}_3 \right) e^{\lambda_1 t}, \quad (21)$$

$$\vdots \quad (22)$$

$$\mathbf{x}_k(t) = \left(\frac{t^{k-1}}{(k-1)!} \mathbf{v}_1 + \frac{t^{k-2}}{(k-2)!} \mathbf{v}_2 + \dots + t\mathbf{v}_{k-1} + \mathbf{v}_k \right) e^{\lambda_1 t}. \quad (23)$$

These k solutions are linearly independent and collectively form a fundamental set of solutions associated with the eigenvalue λ_1 .

- For a repeated eigenvalue λ_1 with algebraic multiplicity k , we need to construct a chain of generalized eigenvectors to generate k linearly independent solutions. The solutions involve terms of increasing polynomial degree multiplied by $e^{\lambda_1 t}$.

Summary: Solution Structure Based on the Nature of Eigenvalues

- The solution strategy for the system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$$

depends critically on the eigenvalues and eigenvectors of the matrix $\mathbf{A}$.

- **Case 1: Distinct Real Eigenvalues**

- If $\mathbf{A}$ has n distinct real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then there exist n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

- The general solution is:

$$\mathbf{x}(t) = \sum_{i=1}^n c_i \mathbf{v}_i e^{\lambda_i t},$$

where $c_1, c_2, \dots, c_n$ are constants.

- **Case 2: Complex Conjugate Eigenvalues**

- If $\mathbf{A}$ has a pair of complex conjugate eigenvalues $\lambda = p + jq$, $\lambda^* = p - jq$, with eigenvector $\mathbf{v} = \mathbf{a} + j\mathbf{b}$, then two real-valued solutions are:

$$\mathbf{x}_1(t) = e^{pt} (\mathbf{a} \cos qt - \mathbf{b} \sin qt),$$

$$\mathbf{x}_2(t) = e^{pt} (\mathbf{a} \sin qt + \mathbf{b} \cos qt).$$

- The general solution involves linear combinations of these two real-valued functions.

- **Case 3: Repeated Eigenvalues**

- If $\mathbf{A}$ has a repeated eigenvalue λ_1 with fewer than n linearly independent eigenvectors, generalized eigenvectors are required.

- Construct a chain of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ satisfying:

$$(\mathbf{A} - \lambda_1 \mathbf{I}) \mathbf{v}_1 = \mathbf{0},$$

$$(\mathbf{A} - \lambda_1 \mathbf{I}) \mathbf{v}_2 = \mathbf{v}_1,$$

$$\vdots$$

$$(\mathbf{A} - \lambda_1 \mathbf{I}) \mathbf{v}_k = \mathbf{v}_{k-1}.$$

- The corresponding solutions are:

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda_1 t},$$

$$\mathbf{x}_2(t) = (t\mathbf{v}_1 + \mathbf{v}_2) e^{\lambda_1 t},$$

$$\mathbf{x}_3(t) = \left(\frac{t^2}{2} \mathbf{v}_1 + t\mathbf{v}_2 + \mathbf{v}_3 \right) e^{\lambda_1 t}, \quad \text{etc.}$$

- These solutions involve polynomial terms multiplied by $e^{\lambda_1 t}$.